

Percentile

A percentile locates a value relative to an ordered distribution.

What does being near the 80th percentile tell us about a file size?

Percentile or percent correct?

A synthetic quiz score of 72 points is at the 90th percentile.

Does this mean the student answered 90% of the questions correctly?

Order first

Synthetic file sizes (KB): 14, 8, 22, 10, 18, 12, 6.

What is their order from smallest to largest?

The median

Synthetic data. The median separates ordered data into lower and upper halves.

What is the median of 6, 8, 10, 12, 14, 18, 22?

An even number of values

Synthetic file sizes (KB): 2, 4, 4, 6, 8, 10, 10, 12.

What is their median?

Quartiles

Q1, Q2, and Q3 mark approximately 25%, 50%, and 75% of an ordered distribution.

Which quartile is the median?

The lower quartile

The first quartile Q_1 is the median of the lower half. Synthetic sizes (KB): 2, 4, 4, 6, 8, 10, 10, 12.

What is Q_1 ?

The upper quartile

The third quartile $Q3$ is the median of the upper half. Synthetic sizes (KB): 2, 4, 4, 6, 8, 10, 10, 12.

What is $Q3$?

Quartiles with an odd sample size

Synthetic ordered values: 1, 2, 3, 4, 5, 6, 20.
Exclude the median when taking medians of the lower and upper halves.

What are Q1 and Q3 under this convention?

Interquartile range

IQR is the distance from Q_1 to Q_3 .

$$\text{IQR} = Q_3 - Q_1$$

For $Q_1 = 4$ KB and $Q_3 = 10$ KB, what is the IQR?

Interpret the middle half

Synthetic file sizes have $Q1 = 4$ KB and $Q3 = 10$ KB.

Approximately what fraction lies between these quartiles?

An outlier rule

A potential outlier lies below $Q1 - 1.5 \text{ IQR}$ or above $Q3 + 1.5 \text{ IQR}$.

For $Q1 = 4$ and $Q3 = 10$, what are the two fences?

Flag a value

Synthetic file sizes have $Q1 = 4$ KB, $Q3 = 10$ KB, and $IQR = 6$ KB.

Does a 30 KB file fall beyond an outlier fence?

At the fence

Synthetic data have $Q1 = 4$ and $Q3 = 10$. The upper fence is 19.

Does the 1.5 IQR rule flag a value equal to 19?

Percentile from a rank

Synthetic ordered values: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20. Use $100 \times (\text{count below} + \text{half the count equal})/n$.

What percentile rank does 14 have under this rule?

A percentile location

Textbook rule: $i = (k/100)(n + 1)$. An integer i selects position i ; otherwise average the values at the adjacent positions.

For $n = 20$ and $k = 60$, which ordered positions determine P_{60} ?

A percentile calculation

Synthetic ordered values: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20. Use $i = (k/100)(n + 1)$; average adjacent positions when i is noninteger.

What is P60?

An integer percentile position

Synthetic ordered values: 3, 5, 6, 8, 10, 13, 16, 20, 25. Use $i = (k/100)(n + 1)$.

What is the 70th percentile?

A percentile from a frequency table

Synthetic values 2, 4, 6, 8 have frequencies 2, 3, 4, 1. Use $i = (k/100)(n + 1)$.

What is P60 without writing out all ten observations?

Percentile rank with ties

Synthetic ordered scores: 2, 4, 4, 4, 6, 8. Use $100 \times (\text{count below} + \text{half the count equal})/n$, rounded to the nearest integer.

What percentile rank does a score of 4 have?

Interpreting a percentile in context

Synthetic service records have a 75th-percentile wait of 12 minutes.

What does this say about waits on either side of 12 minutes?

Ties in percentile statements

Half of a synthetic class has the same quiz score as the median.

Why should we say approximately half rather than exactly half score below the median?

Location and spread

Two synthetic datasets have the same median but IQRs of 4 KB and 20 KB.

What difference does the IQR reveal?